

Assignment # 4

Subject: Advanced Software Engineering (Adv SE)



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Section:

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Q1. Pareto Data Table & Calculations

First, arrange all software defects in descending order according to frequency.

Total Bugs Reported = 200

Step 1: Sorted Defect Data

1. Buffering & playback freezes = 90
2. Subscription/Payment processing errors = 50
3. Audio-video sync delays = 30
4. Subtitles failing to load = 20
5. Profile picture upload errors = 10

Step 2: Cumulative Frequency & Percentage Calculations

Buffering & playback freezes

- Frequency = 90
- Cumulative Frequency = 90
- Cumulative Percentage = $(90 \div 200) \times 100 = 45\%$

Subscription/Payment processing errors

- Frequency = 50
- Cumulative Frequency = $90 + 50 = 140$
- Cumulative Percentage = $(140 \div 200) \times 100 = 70\%$

Audio-video sync delays

- Frequency = 30
- Cumulative Frequency = $140 + 30 = 170$
- Cumulative Percentage = $(170 \div 200) \times 100 = 85\%$

Subtitles failing to load

- Frequency = 20
- Cumulative Frequency = $170 + 20 = 190$
- Cumulative Percentage = $(190 \div 200) \times 100 = 95\%$

Profile picture upload errors

- Frequency = 10
- Cumulative Frequency = $190 + 10 = 200$

- Cumulative Percentage = $(200 \div 200) \times 100 = 100\%$

Software Defect Type	Frequency	Cumulative Frequency	Cumulative Percentage
Buffering & playback freezes	90	90	45%
Subscription/Payment processing errors	50	140	70%
Audio-video sync delays	30	170	85%
Subtitles failing to load	20	190	95%
Profile picture upload errors	10	200	100%

Q2. Applying the 80/20 Rule

According to the Pareto principle, a small number of defects are usually responsible for the majority of system problems.

In this case, the following defects form the “vital few”:

- Buffering & playback freezes
- Subscription/Payment processing errors
- Audio-video sync delays

Together, these issues contribute to approximately 85% of the total user complaints.

Justification

Management should focus on these defects first because they affect the largest number of users and have the biggest impact on customer satisfaction. Buffering and playback issues directly disturb the viewing experience, while payment errors can lead to financial loss and reduced customer trust. Audio-video sync problems also make content difficult to watch properly.

By solving these high-frequency defects first, the company can reduce the majority of complaints quickly instead of wasting time and resources equally on smaller issues with lower impact.

Q3. Fishbone / Ishikawa Root Cause Analysis

Selected Defect:

Buffering & playback freezes (Highest frequency = 90)

Possible Root Causes under 6M Categories

Man (Developer/Team Skills)

Developers lacked experience in optimizing video streaming performance and caching techniques.

Machine (Hardware/Server/Testing Environment)

Streaming servers had limited bandwidth and became overloaded during peak traffic hours.

Method (Processes/Development Practices)

Performance testing was not properly conducted before deploying the updated codebase.

Material (Code Dependencies/APIs/Libraries)

An outdated media streaming library caused inefficient video data handling.

Measurement (QA Metrics/Testing Thresholds)

The QA team focused more on functionality testing and ignored network latency benchmarks.

Mother Nature (Environmental Factors)

Users with unstable internet connections experienced higher buffering problems during bad weather or network congestion.

Q4. Corrective Quality Action Plan**1. Buffering & Playback Freezes**

The development team should implement load balancing and optimize video caching mechanisms. In addition, automated performance testing should be added before every deployment to identify playback issues early.

2. Subscription/Payment Processing Errors

The developers should add stronger API validation and exception handling for payment gateways. Automated transaction testing should also be performed regularly to ensure secure and reliable payment processing in future software builds.